SentinelMesh

An Integrated Trust Fabric for Securing Multi-Agent LLM Systems

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One-line pitch

A single security platform that verifies an AI model is *safe before you load it*, verifies an AI agent is *who it claims to be while it's running*, verifies the *instructions it's acting on haven't been injected*, and uses AI to watch the *traditional infrastructure* those agents run on — genuinely two-way: AI protecting AI, and AI protecting everything AI depends on, fused into one risk score.

Why this stands out

Right now, if a company wants to secure an LLM-agent deployment, they need three separate tools from three separate vendors — a model-scanning product, an identity/access product, and a prompt-injection firewall — none of which talk to each other or share a common risk signal, and none of which look at the actual servers/network the agents run on. Recent surveys explicitly call this out as a gap: multi-agent and supply-chain threats are under-studied relative to prompt injection alone, and no open system currently unifies all three under one architecture. SentinelMesh closes that gap **and** deliberately balances both directions of AI-security research that usually stay separate:

- **Security for AI** (AI is the thing being protected) — WeightWatch, GhostTrace, Prompt-Injection Defense
- **AI for security** (AI is the method protecting traditional infrastructure) — SentinelGuard, the new infra/log anomaly-detection module below

Both directions feed the same Unified Risk Engine, so the "AI defends AI" and "AI defends infrastructure" halves of the project are genuinely hand-in-hand, not two separate projects bolted together. This is closer to how a real security company (e.g. an "AI-native CrowdStrike") would actually design this, than a single-technique research demo.

System Overview — Five Modules

Module 1 — WeightWatch (Supply-Chain Scanner)

Static, pre-deployment scanner for backdoored models and malicious LoRA/PEFT adapters. Analyzes weight-space statistics (singular value concentration, entropy, distribution shape) without executing the model — so it's fast and safe to run on untrusted downloads.

- **Adds cybersecurity depth:** cryptographic hash-verification and signing of "cleared" models (so a scanned-and-passed model can be trust-chained), integration with a community-style registry of known-malicious model hashes (threat intel feed pattern).
- **Adds AI/ML depth:** trained classifier over weight-space features; SHAP-based explainability so a verdict isn't a black box ("flagged because of anomalous singular-

value entropy in layer 14").

Module 2 — GhostTrace (Runtime Identity & Trust Engine)

Behavioral fingerprinting of each agent in a multi-agent system (tool-usage patterns, API-call sequences, communication style) feeding a continuously updated trust score, catching impersonation and rogue delegation as it happens rather than only at login.

- **Adds cybersecurity depth:** cryptographically signed inter-agent messages, a communication graph of who's talking to whom, incident-forensics timeline for compromised agents.
- **Adds AI/ML depth:** a Graph Neural Network over the agent communication graph (not just per-agent behavior, but who's connected to whom) — this is the strongest "novel" lever in the whole project, since GNN-based trust modeling for AI agents specifically is barely explored yet.

Module 3 — Prompt-Injection Defense

Adaptive classifier/guardrail against direct and indirect prompt injection, continuously re-evaluated against an evolving, team-generated attack corpus rather than a static benchmark (following the finding that static-benchmark defenses get broken by adaptive attackers).

- **Adds cybersecurity depth:** red-team attack-crafting methodology, MITRE ATLAS technique mapping for every detected attack (so output is analyst-readable, not just "flagged/not flagged").
- **Adds AI/ML depth:** the guardrail model itself, plus an adversarial training loop where the classifier retrains against its own red team's newest evasion attempts each cycle.

Module 4 — SentinelGuard (AI-for-Security: Infrastructure Anomaly Detection)

This is the module that flips the direction: instead of AI defending an AI asset, this is AI defending the traditional infrastructure the agents run on — the servers, network, and logs. A Graph Neural Network models hosts, processes, and network flows as a graph and flags anomalous connection patterns and log sequences, following the established GNN-based intrusion-detection line of work (e.g. E-GraphSAGE, Lo et al., IEEE/IFIP NOMS 2021; and the broader line of GNN-IDS research surveyed through 2026).

- **Adds cybersecurity depth:** realistic infrastructure attack simulation (lateral movement, credential abuse, exfiltration) generated against the same testbed hosting the agents — directly reusable from prior Active Directory attack/defense experience on the team.
- **Adds AI/ML depth:** graph-based anomaly detection over host/network/log telemetry, trained and evaluated against the simulated attacks, with the same explainability standard (SHAP or GNNExplainer-style attribution) as the other modules.

Module 5 — Unified Risk Engine & Dashboard (the integration layer)

This is what makes it one product instead of five, and where the two directions of the project — AI defending AI, and AI defending infrastructure — genuinely converge. A single

backend fuses signals from all four defense modules into one composite risk score per agent/session, displayed on a live dashboard (trust scores, scan verdicts, injection alerts, infra anomaly alerts, communication graph) — the equivalent of a SOC console, but purpose-built for AI-agent ecosystems and the infrastructure beneath them.

- **Adds cybersecurity depth:** SIEM-style event logging and alerting conventions, human-override/audit trail for every automated decision.
- **Adds AI/ML depth:** the risk-fusion model itself — a calibrated ensemble that weighs four independent signals (two "security-for-AI," one "AI-for-security," plus identity) into one number, which is a genuinely non-trivial ML problem (not just averaging four scores).

Task Division (0 → End, by person)

Teammate 1 — Cybersecurity: Supply-Chain & Infrastructure Attacks

- Own the threat modeling and literature review for Module 1 (WeightWatch) and Module 4 (SentinelGuard).
- Craft poisoned LoRA/PEFT adapters with varied trigger designs; run evasion attempts against the WeightWatch classifier as it matures.
- Design the cryptographic signing/hash-verification scheme for "cleared" models.
- Simulate realistic infrastructure attacks (lateral movement, credential abuse, exfiltration) against the testbed hosting the agents, to generate attack traffic for SentinelGuard — directly reusable from prior AD attack/defense experience.
- Own the incident-forensics/attack-timeline feature of the dashboard.
- Co-own final report sections on supply-chain and infrastructure threat models.

Teammate 2 — Cybersecurity: Identity, Injection & Threat Taxonomy

- Own the threat modeling and literature review for Module 2 (GhostTrace) and Module 3 (Prompt-Injection Defense).
- Build impersonation and rogue-delegation attack scenarios against the multi-agent testbed for GhostTrace.
- Build and continuously evolve the direct/indirect prompt-injection attack corpus; run the red-team side of the adversarial training loop.
- Map every detected attack across **all four modules** — both MITRE ATLAS (AI-specific) and MITRE ATT&CK (traditional infra) technique IDs — into one shared taxonomy for analyst-readable output. This is the piece that makes the "AI-for-security + security-for-AI" split legible in one place, not just architecturally.
- Design the cryptographically signed inter-agent messaging protocol used by GhostTrace, and own SIEM-style event logging conventions / audit-trail feature.
- Co-own final report sections on identity/injection threat models and the unified MITRE mapping.

Teammate 3 — AI/ML: Supply-Chain Model & Infrastructure Anomaly Detection

- Build the weight-space feature extraction pipeline and classifier for WeightWatch (security-for-AI); retrain against Teammate 1's evasion attempts.
- Implement SHAP-based explainability for scanner verdicts.
- Build the GNN-based infrastructure anomaly-detection model for SentinelGuard (AI-for-security), trained against Teammate 1's simulated attacks.
- Co-lead the Unified Risk Engine with Teammate 4 — specifically owns how the two "AI for/of security" signals (WeightWatch + SentinelGuard) feed into the fusion model.
- Co-own final report sections on the scanner model and infrastructure anomaly detection.

Teammate 4 — AI/ML: Identity Modeling, Injection Defense & Risk Fusion

- Build the behavioral fingerprinting pipeline and the GNN-based trust-scoring model for GhostTrace (the project's strongest novelty lever).
- Build the prompt-injection guardrail classifier; own the adversarial training loop against Teammate 2's evolving attack corpus.
- Co-lead the Unified Risk Engine with Teammate 3 — specifically owns how the two "security-for-AI" signals (GhostTrace + injection defense) feed into the fusion model, and owns the live dashboard frontend.
- Co-own final report sections on GNN trust modeling, injection defense, and risk fusion.

Shared across all four (not owned by one person)

- Initial literature review and threat-model workshop before any building starts.
- Weekly cross-module sync so attack/defense pairs stay in lockstep (e.g., Teammate 1's adapter and infra attacks vs. Teammate 3's scanner and anomaly detector; Teammate 2's identity/injection attacks vs. Teammate 4's trust model and guardrail).
- Joint design session for the Unified Risk Engine (Teammate 3 + Teammate 4) so the fusion model is built together, not handed off — this is the literal "hand in hand" point of the whole system.
- Recruiting and onboarding real external users once the system is deployable.
- Collecting and triaging user feedback, then re-assigning refinement work per module owner.
- Final report writing, integration testing, and demo preparation.

Why this division works

Every cybersecurity teammate owns attack surfaces end-to-end and stays paired with the specific teammate whose module their work stress-tests, so the "red team vs blue team" loop is real and continuous. Splitting AI-for-security (SentinelGuard) and security-for-AI (WeightWatch, GhostTrace, Injection Defense) across both AI/ML teammates — rather than giving one person all of one direction — means the Unified Risk Engine is genuinely co-owned at the point where both directions meet, instead of one person integrating

someone else's work. Each of the four also gets at least one clearly novel, resume-worthy contribution: cryptographic model trust-chaining + infra attack simulation (Teammate 1), a unified AI/traditional threat taxonomy (Teammate 2), explainable supply-chain and infrastructure anomaly detection (Teammate 3), and GNN-based agent trust modeling plus risk fusion (Teammate 4) — so individual contribution sections in the final report and on each person's resume are distinct and substantial, not interchangeable.